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Analysis of Block Chain Applications in Asset Backed Securitization System

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Abstract

Blockchain is an emerging technology penetrating various real-time and sensitive domains including financial products. Block chain can deal with complex system of financial transactions with focus to provide security and integrity to both the parties with reliable flow of transactions and asset securitization. Block chain is a distributed technology providing a lot of benefits such as cryptic transactions, smart contracts, distributed ledger system along with decentralization, authentication, authorization, and immutability to enhance the efficiency of the financial system. In this paper, the existing solutions for asset-backed securitization in blockchain are analysed and reviewed to highlight the requirements for change in design and ideas for enterprise applicability of blockchain for asset securitization. Also, the benefits of blockchain to reduce the risk.

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Keywords: Blockchain; Securitization; Decentralization; Finance

1. Introduction

Blockchain was invented in 2008 and provided an efficient base for various real-time domains such as supply chain management [1], stocks management [2], healthcare [3] and transaction management [4]. Apart from this, blockchain has the capability to optimize the business processes and minimize operation costs by removing the intermediaries. Blockchain technology supports to develop a decentralized and innovative system for improving the efficiency of the defined processes [5].

Blockchain is a decentralized technology that utilizes cryptographic techniques for secure recording of transactions or information in a distributed network of computers. In Blockchain, all committed transactions are stored in a chain of blocks. Each block within the blockchain is identified by a hash, generated using the SHA256 cryptographic hash

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algorithm [6]. The generated hash is random in nature and a minute change in the block changes the entire hash of the block. Hence, it is not feasible for anyone to manipulate the data because it would require changing all the blocks in the chain, which is nearly impossible.

In financial transactions, blockchain can greatly save the cost as well as it increases the efficiency. Blockchain is a decentralized system which allows users to make payments directly without any involvement of third parties such as banks. This will not only reduce the transaction cost and also make it safe and secure by eliminating the role of intermediaries. Due to this blockchain can be used in various financial services such as digital assets, online payment and many more. Blockchain has provided numerous benefits to the banking system by introducing security, transparency, and trust. These systems with the help of block chain provide high reliability and clarity to ensure best services to customers. Blockchain's application value is reflected in its ability to establish a relatively reliable information screening and anti-counterfeiting mechanism [7][8]. Overall, blockchain technology can easily transform many industries. But it is still in its early stage and many challenges need to be met before reaching its full potential. Artificial intelligence can also be integrated with block chain to explore its applications in computer vision and healthcare [9]-[11].

The key contributions of this paper are: (1) to identify the key terminologies of the blockchain and their utilization in financial systems, (2) determine the role of blockchain in financial assets securitization and (3) review the current work in the domain of assets securitization using block to provide solutions to the existing issues.

The rest of the paper is designed as follows: Section 2 elaborates the basic terminologies in the blockchain. The discussion about the financial assets securitization is incorporated in the section 3. Section 4 reviewed the existing work in blockchain for financial assets securitization. Lastly, the concluding remarks are sketched in the Section 5.

2. Blockchain basic terminologies

Blockchain technology employs mechanism to make the financial transaction secure and reliable. Block chain utilizes cryptographic techniques, consensus mechanism, and distributed ledge for the transparency and integrity of the financial assets securitization. The details of the basic architecture of the blockchain is as follows.

2.1 Blockchain design architecture

Blockchain architecture is the technical design and structure of a blockchain network, it is a distributed and decentralized digital ledger that that keep the records of the transactions between parties in a transparent manner. The various components that make up a blockchain architecture are illustrated in Fig. 1.

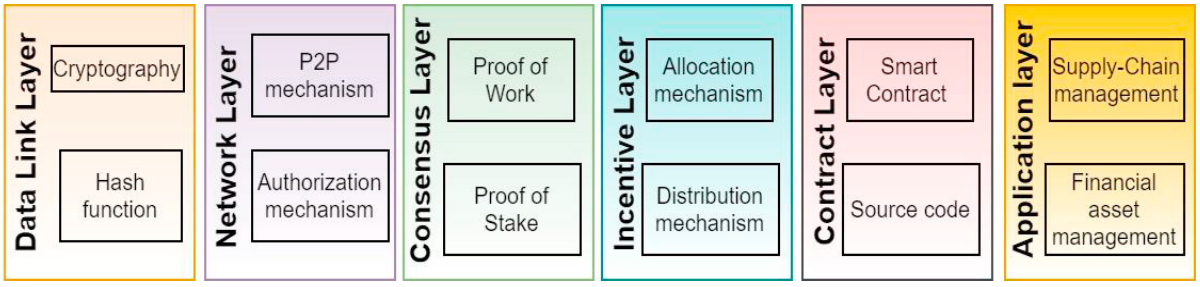


Fig. 1. Various components of blockchain architecture

The basic terminology used in blockchain description is as follows:

1.*Nodes:* These are network stakeholders or devices that participate in the network by keeping track of the blockchain

and verifying transactions.
2.*Transactions:* These are records of data that are added to the blockchain, it includes things like financial transactions of digital assets.
3.*Blocks:* these are like data structures within database of blockchain. Block records the recent transactions that are verified and added to the blockchain in a sequential order.
4.*Consensus Mechanisms:* Blockchain works on set of rules that guide how the network reaches agreement after the validity of transactions. This is a copy of terms and conditions on which all the untrusted parties agreed on, before to begin with the flow of information.
5.*Cryptography:* It is the mathematical algorithms and encryption techniques of securing data from unauthorized access and maintain integrity of the blockchain network. The hash code for each block is computed for each block of block chain.
6.*Smart Contracts:* These Smart Contract ensures that both parties to the contract cannot tamper with the content of the contract and that the contract can be enforced without the supervision of a central administrator. Its code can automatically execute and track the terms and conditions of the agreement. Thus, Smart Contracts are self-verifying, self-executing and irreversible [12].

2.2 Description of hash algorithm

Hash algorithm collects data from the user and makes it a fixed sized output called a hash value which is unreadable. Generally, hash algorithm is a one-way function process. It means that plain text can not be recovered from the hashed text. Hash algorithms has wide applications in security the data to ensure data integrity, authentication, and non-repudiation. For example, company A uses hash algorithm to create a digital fingerprint for security of asset tokens that can be used to verify its authenticity and detect any modifications or tampering. The most used hash algorithm is SHA-256 (Secure Hash Algorithm 256-bit) in blockchain. Hash algorithms have an imperative role in securitization by providing a secure way to verify the integrity of data. Fig. 2 illustrates the process of hashed text generation using hash algorithms.

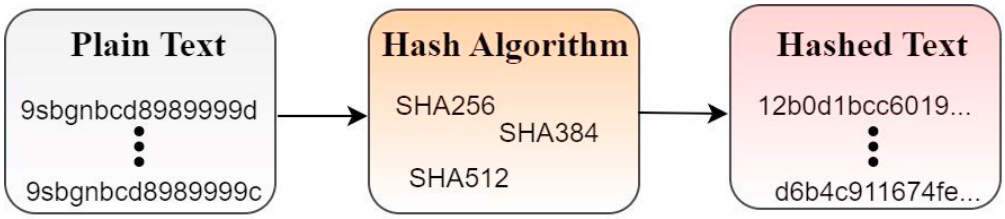


Fig. 2. Description of Hash Algorithm in hashed text generation.

2.3 Consensus mechanism

Consensus mechanisms are used for transaction verification in block chain for financial asset securitization. It is utilized to secure the agreement among the transacting nodes in the blockchain network. When the network expands, blockchain demands all the new users to verify themselves for the security and authenticity of the node's transactions. Consensus mechanism is dynamic in nature and highly depends on the transaction type and environment. Based on this, consensus mechanism can be categorized as primitive, alternative and extension [13]. The commonly used types of consensus mechanism are proof of work (PoW) and proof of stake (PoS) for Bitcoin and Peercoin, respectively.

2.4 Smart contract

Smart contract is an innovative practice in blockchain to automate the cross organizational business processes. Smart contract in block chain provides a unified architecture using decentralized autonomous organization (DAO) for financial transaction management [14]. DAO in blockchain is an effective way for encoding financial transactions through smart contracts. Smart contracts are sharable documents between multiple untrusted transacting parties to ensure transparency and integrity among them. It incorporates the suggestions from all the involved parties before execution of the transaction. Smart contracts can be either one-to-many or many-to-many. Former is existing between the one party who deploy the contract and all the other parties on the blockchain while in the later multiple parties are

involved.

2.5 Distributed ledger

This technique is used to make the block chain decentralized by removing the necessity of central authority. Distributed leger maintains a digital database to share, validate, authenticate, update, and synchronize the nodes in the networks to ensure transparency and integrity. Distributed leger can be public or private depending on the type of block chain network. All the data in the distributed ledger is secure, permanent, and safe. The smart contracts are merged into distributed legers to improve reliability and transparency of financial transactions applications.

2.6 Block chain types

Typically, the blockchain networks are categorized into 4 main types [15].

Public Blockchain: These blockchain networks are open to the public and hence, are low on efficiency. There is no restriction on entry and exit of the participants in the block chain network. The most popular example of public block chain is bitcoin.

Private Blockchain: This type of blockchain is designed for a specific group of users such as business or organization and with limited network access. This network is for internal purposes and has permissioned consensus process. Hyperledger Fabric is commonly implemented application of a private blockchain.

Consortium Blockchain: This type of block chain network includes a group of private blockchain network, owned and operated by a consortium or a group of organizations that work together to maintain the network. Example of consortium network is Hyperledger.

Hybrid Blockchain: This blockchain integrates the benefits of both public as well as private blockchains. It allows the public participation up to a certain level and restricts access to specific parts of the network too. The example of hybrid block chain is Ripple.

3. Overview of financial assets securitization

Securitization is a process of pooling various financial assets such as consumer loans, mortgages by the originator etc. These assets are transferred to Special Purpose Vehicles (SPVs). These SPVs issue the securities to investors in bonds. The securities can be categorized either as Mortgage-backed Securities (MBS) or Asset-backed Securities (ABS). Generally, securitization is related to create new securities which are backed by the cash flows from the assets. The securities created through securitization are divided into different tranches, each with their own level of risk and return. The main benefit of securitization increases liquidity by selling the assets to investors. The originator or bank can free up capital and reduce its risks of those assets. However, securitization is complex structure and can be associated with high risk. Overall, securitization is a financial process that can provide benefits to both investors and originators of financial assets but requires careful consideration and risk management.

3.1 Usage of Block chain for Asset backed securitization

Asset securitization is a complex structured financial product provides numerous advantages with block chain technology. Important aspects in securitization include originators, servicers, rating offices, trustees as well as financial experts can be impacted by the blockchain to simplify processes, cost reduction, speed up exchanges, improve transparency and enhance security. The following are the features of block chain incorporation into asset backed securitization.

i) Improved transparency

The originator registers a new blockchain in system, incorporate asset data and broadcasts it to all the nodes connected in the network. In the liquidation process of ABS, multiple services are provided for the asset. It must be ensured that the assets are in accordance to accounting laws, specify credit rating and other specific project information before broadcasting it to all the blocks in the block chain. Apart from the data blocks, block chain includes legal, regulatory and evaluation blocks too. These additional blocks are used for making the information reliable, transparent, and unique. User access rights are also defined to view, change, and update the asset and the trading information any times. The block chain ensures opaqueness in the networks [4]. Once the data is stored in blockchain there is no need to

repeat the work. Hence by creating a secure and immutable records of the transactions confidence of the investors increases.

ii) Enhanced network throughput
Integration of blockchain into ABS business removes the involvement of multiple intermediates in the network. It can reduce human and material costs to large extent [16]. Smart contract is a set of code stored on a blockchain network. It is executed automatically when certain conditions are met. Once a smart contract is deployed on a blockchain, it cannot be breached. Smart contracts easily eliminate the work of intermediaries, or we can say the intermediates. So basically, smart contracts eliminate the time-consuming manual verification in ABS and makes it automatic. Transactions are now reported and regulated in real time without brokers.

iii) Tokenization of assets
Token is a digital part of asset. Tokenisation is the process of converting real asset into digitally tradeable asset. Blockchain creates tokens of securities. These tokens are cheaper, faster, more transparent as compared to the current trading of securities [18]. Tokens settlements can be completed within minutes compared to the current practice of T+2 days. Tokens trading is possible 24*7 supported by smart contracts which eliminate the settlement delay. Tokenised asset is equivalent to be listed as other shares are listed in share market.

iv) Asset management
The blockchain has the capability to monitor and maintain the changes in the assets. Blockchain itself process the assets and complete the exchange of tokens. The payment records of investors and investors are recorded digitally [17]. As now the credit rating department used to give credit rating which is not fully transparent but now the blockchain algorithm will give transparent as well as more accurate rating to securities.

3.2 Critical analysis of block chain with asset backed securitization

The main functionality of the block chain with ABS includes streamlining the securitization process by incorporating secured path flow, reduced delays, enhanced transparency and improved accessibility. Block chain offers security using cryptographic generated hash function of the transaction blocks, decentralized ledger, smart contracts to reduce the settlement time of the transactions and immutable record of transactions. The integration of block chain with ABS offers interoperating, faster and transparent system to trace sensitive transactions.
Apart from this, there are certain limitations of integrating the block chain with ABS. First, there is a requirement of huge costs for migrating the system to prepare it for integration. Block chain is a recent and distributed technology, whose capability for secure financial transactions are yet to be tested. The distribution of sensitive information to all the networks nodes may lead to security breach. Another bottleneck is from the stakeholders and the policy makers who all are to be convinced for the migration of the system on the reliable technology like block chain.

4. Existing work in securitization using block chain

Table 1 tabulates the recent work utilizing blockchain on ABS. Also, the detailed analysis and limitation of the published work are elaborated.

Table1: Related work in the domain of ABS with Blockchain

SNo	Reference/Authors	Year	Description	Limitations	Summary
1.	Gupta et al. [15]	2021	Transactional methods integrating securitization and distributed ledger	High transactional cost and less security	Temper proof system with removal of third untrusted parties to ensure security and integrity.
2.	Pan and Qiu [16]	2020	ABS China financial system	Security issues and asset management issues	Integrate block chain with ABS to address the limitation of conventional ABS.

3.	Wan et al. [17]	2017	ABS for customer credit line for Internet financial firms	Lack of asset liquidity with restricted asset operations.	Highlights the concept of accounts receivable BS, ABS cloud platform and necessity of blockchain in ABS.
4.	Sun [18]	2022	Accounts receivable blockchain securitization	Stability of cash flow with inefficient management and asymmetric information	Explore the receivable securitization in block chain for transparency and openness.
5.	Zhang et al. [19]	2018	Privacy-preserving and user-controlled data sharing architecture	Scalability, security, and confidentiality issues.	Fine-grained access control, based on the blockchain model and attribute-based cryptosystem.
6.	Hongo and Jing [20]	2018	Security asset transaction based on peer-to-peer networks	Buying and selling of assets computationally costly.	Attribute based selection of user basic information for financial transaction management.
7.	Foglie et al. [21]	2021	Financial market digitization standardizing smart contract and asset tokenization	Tangible and non-tangible assets boundless distribution	Smart Contract use and asset tokenisation on the distributed ledger.

5. Conclusion

In this paper, we have analysed the various block chain-based ABS platforms for solving the limitations of the financial systems. We have summarized the limitations of the conventional ABS system and highlights the benefits of blockchain such as integrity, security, transparency and reliability. It has been evident that the incorporation of block chain in ABS will not only benefits the investors and the participants but also provide a tamper-proof and decentralized securitization system.

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